

Organisms & Populations

Topic - Abiotic Factors, Responses to Abiotic factor, Adaptation

1. What is the correct order of ecological hierarchy:-

(1) Species → Genes → Family → Order → Class → Division → Kingdom

(2) Species → Community → Order → Kingdom

(3) Organism → Population → Community → Biome

(4) Organism → Community → Biosphere → Ecosystem → Biome

2. Read the statements and find how many are incorrect with respect to adaptation to temperature and pressure tolerance

(a) Antarctic fish survives below 0°C as their body contains antifreeze solute

(b) Marine invertebrates and fish adapted biochemically to survive at greater depth

(c) Physical work is needed to avoid altitude sickness

(d) Xerophytes have thick cuticle to mitigate water loss

(1) 4

(2) 3

(3) 2

(4) 1

3. Who is the Indian father of Ecology?

(1) E. Hackel

(2) Ramdeo Mishra

(3) P. Odum

(4) Tansley

4. At high altitude we feel the sickness. The reason for sickness may be due to

(1) Low atmospheric pressure

(2) High atmospheric pressure

(3) High temperature

(4) Low temperature

5. Given below are two statements:

Statement-I : All autotrophic plants are self-sustainable and do not need any other species for survival.

Statement-II : In nature animals, plants and microbes do not and cannot live in isolation but interact in various ways to form biological community.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

6. Snow leopard are not found in _____ and tuna fish rarely caught beyond _____ latitudes in the ocean.

(1) Tropical, Kerala

(2) Kerala, tropical

(3) Kerala, temperate

(4) Kerala, sub-tropical



7. Kangaroo rat in _____ American deserts is capable to meet all its water requirement through _____.

- (1) East, internal fat oxidation
- (2) North, internal fat oxidation
- (3) North, internal protein oxidation
- (4) West, internal fat oxidation

8. Smaller animals tend to lose body heat very fast as compared to larger animals because they have

- (1) Lower surface area to volume ratio
- (2) High surface area to volume ratio
- (3) Equal surface area and volume
- (4) Low based metabolic rate (BMR)

9. Natural selection operates at

- (1) Organismal level
- (2) Population level
- (3) Community level
- (4) Ecosystem level

10. Some species are capable of burrowing into the soil to hide and escape from the above ground heat. Such adaptations are referred to as

- (1) Physiological adaptations
- (2) Morphological adaptations
- (3) Behavioural adaptations
- (4) Biochemical adaptations

11. Stage of suspended development called

- (1) Diapause
- (2) Aestivation
- (3) Hibernation
- (4) Migration

12. Warm blooded animals have larger body size in cold zone. This is called

- (1) Allen's rule
- (2) Jordan's rule
- (3) Rensch's rule
- (4) Bergmann's rule

13. Study the following statement and select the correct ones. –

i) Organisms capable to tolerate a wide range of temperature are called stenothermal organisms.

ii) Thermal tolerance of different species determines their geographical distribution to a large extent.

iii) Average temperature in tropical desert in summer is $<50^{\circ}\text{C}$.

iv) Thermal spring cannot sustain life due to very high avg. temperature i.e. $>100^{\circ}\text{C}$.

- (1) ii
- (2) i, iii, iv
- (3) i, ii, iv
- (4) iv

14. Read the statements carefully and choose the correct option

(A) Hekistotherms are restricted to alpine zone

(B) Organisms are adapted to brief summer of below 10°C and long snowy winter

- (1) Both are correct
- (2) Only B is correct
- (3) Only A is correct
- (4) Both are incorrect

15. Match the columns

Column-I	Column-II
(a) Hibernation	(i) Bear
(b) Diapause	(ii) Plants
(c) Migrate	(iii) Zooplanktons
(d) Conform	(iv) Siberian crane

(1) (a) → (i), (b) → (iii), (c) → (iv), (d) → (ii)

(2) (a) → (i), (b) → (iii), (c) → (ii), (d) → (iv)

(3) (a) → (ii), (b) → (i), (c) → (iii), (d) → (iv)

(4) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

16. Organisms reacted to narrow range of temperature are called as

- (1) Eurythermal
- (2) Stenothermal
- (3) Heterothermal
- (4) Stenohaline

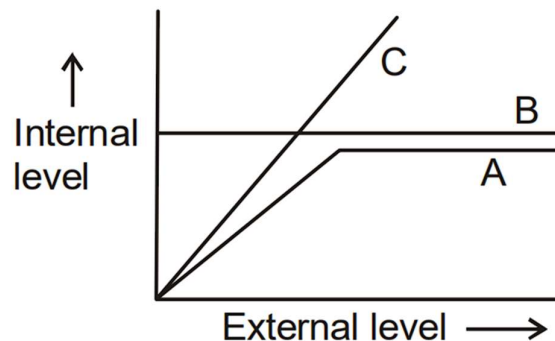
17. Thermoregulation is energetically expensive process for small animals due to their-

- (1) Small surface area relative to their size
- (2) Large size relative to surface area
- (3) Both B and A
- (4) Large surface area relative to size

18. Winter sleep is known as _____ and summer sleep is known as _____.

- (1) Hibernation, Aestivation
- (2) Migration, Aestivation
- (3) Aestivation, Hibernation
- (4) Aestivation, Migration

19. Identify the lines present in the given graph A, B and C



Choose the most appropriate answer from the options given below :

(1) A - Partial regulators B – Regulators C - Endotherms

(2) A - Partial regulators B – Ectotherms C – Endotherms

(3) A - Partial regulators B – Regulators C – Conformers

(4) A – Conformers B – Ectotherms C - Partial regulators

20. Read the following statements

(a) Success of mammals is due to their ability for homeostasis.

(b) Very large animals are rarely found in polar regions.

(c) Nearly all plants can maintain a constant internal environment.

(d) A stage of temporary movement from stressful habitat is aestivation

In the light of the above statements, choose the incorrect statements from the options given below :

- (1) One
- (2) Three
- (3) Two
- (4) Four

ANSWER KEY

1. (3) 2. (4) 3. (2) 4. (1) 5. (3) 6. (2) 7. (2) 8. (2) 9. (2) 10. (3)
11. (1) 12. (4) 13. (1) 14. (1) 15. (1) 16. (2) 17. (4) 18. (1) 19. (3) 20. (2)

